

A Multiple-Radix MAP-Decoder Microarchitecture and Its ASIC Implementation for Energy-Efficient and Variable-Throughput Applications

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Abstract—This article proposes reconfigurable maximum *a posteriori* (MAP) decoding algorithm which operates in multiple radix modes to hard decode the information bits with variable throughput by consuming constant power. Subsequently, a new digital decoder-architecture for this MAP decoding algorithm has been designed that operates in radix-2/4/8 modes. Furthermore, reconfigurable microarchitectures of state-metric and logarithmic-likelihood-ratio (LLR) computation units are presented in this article. Performance analyses of the proposed algorithm have been performed in additive-white Gaussian-noise (AWGN) channel environment where it achieved a bit-error-rate (BER) of 10^{-4} at a signal-to-noise ratio (SNR) of 5 dB. Suggested multiple-radix MAP-decoder is fabricated in united-microelectronics-corporation (UMC) 130-nm-CMOS technology node and it occupies die and core areas of 2.35 and 1.28 mm², respectively, operating at a maximum clock frequency of 204 MHz. This MAP decoder ASIC-chip delivers throughputs of 201, 403, and 605 Mb/s while operating in radix-2, 4 and 8 modes, respectively, consuming a total power of 92 mW. Our chip achieves 0.15 nJ/bit of energy efficiency, which is 30% better than the most energy-efficient implementation from literature. Likewise, it delivers 39% higher throughput and 70× better area-efficiency than the state-of-the-art work.

Index Terms—Application-specific integrated circuits (ASIC) implementation, channel encoder/decoder, CMOS, digital architectures, maximum *a posteriori* (MAP) decoding algorithm, wireless communication system.

I. INTRODUCTION

TURBO and low-density parity-check (LDPC) codes are exceptionally powerful forward-error-correcting codes that deliver near Shannon-limit performance [1], [2]. These channel codes are widely adopted by commercial wireless-communication standards as well as satellite communication applications. Specifically, turbo code has been used in the physical layers of various wireless standards like WCDMA, DVB-SH, HSDPA, HSDPA+, third-generation partnership-project (3GPP)-LTE, and 4G-LTE-Advanced [3], [4]. Similarly, LDPC codes are used by IEEE 802.11n (WiFi), IEEE 802.16e (WiMAX), HPAV, DVB-S2, DVB-T2, DVB-RCS,

ITU-T G.hn, CMMB, and DTMB wireless standards [5], [6]. An imperative processing engine of turbo/LDPC decoder is the MAP decoder that decodes convolutional code based on Bahl–Cocke Jelinek–Raviv (BCJR) algorithm [7]. Hence, the decoding performance, throughput, power consumption, and latency of such turbo/LDPC decoder primarily depend on the design of MAP decoder architecture [8]. Various contributions have been reported that claim to improve such design parameters of MAP decoder. A block-interleaved pipelined architecture that allows the pipelining of add-compare-select kernel of MAP decoder has been proposed by Lee *et al.* [9]. Tang *et al.* [10] designed radix-16 architecture with 2-D add-compare-select and further applied retiming technique to enhance the throughput. An area-efficient MAP processor that supports both single and double binary decoding for high-throughput application is reported by Lin *et al.* [11]. Design and implementations of various MAP-decoder architectures that support different trellis states are comprehensively presented by Studer *et al.* [12]. Shrestha and Pailly [13] proposed a high throughput MAP decoder-architecture based on ungrouped backward recursion technique that enabled it to be deeply pipelined. A lossless parallel implementation of MAP decoder in graphics processing unit (GPU) platform for low-latency application has been reported by Natarajan and Chandrachoodan [14]. Recently, a highly hardware-resource shared architecture of MAP decoder, that alleviates its global routing complexity, has been proposed by Lin and Hsieh [15]. Subsequently, a new MAP decoding algorithm has been proposed by Le *et al.* [16] that is specifically compliant to higher-radix turbo-decoding algorithm.

Recently, 3GPP standardized LDPC code for three use cases of 5G new-radio (NR) technology [17], [18]: enhanced mobile broadband (eMBB) that addresses data-driven cases of accessing multimedia content, massive machine type communication (mMTC) which is characterized by large volume of connected devices (like IoT applications) with low cost plus longer battery life, and ultrareliable low-latency communications (URLLCs) that targets the delivery of critical communication [19]–[21]. Due to proliferation of data usage, 5G-NR standard has specified the peak data-rate/throughput of 20 Gb/s [18]. However, such peak throughput must be supported only by eMBB use-case [17]. Unlike, the throughput requirements of mMTC and URLLC use-cases are lower than 20 Gb/s [18]. Therefore, MAP decoder architecture

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(i.e., fundamental unit of LDPC decoder) must support variable throughput ranging from minimum to peak values. The key design challenge is to conceive an MAP decoder-architecture that meets such variable throughput requirement in an energy efficient manner for better battery life of handheld wireless devices. Fundamentally, throughput delivered by implemented decoder can be varied from lower to higher value using two approaches: increasing the operating clock frequency and adopting parallel decoder-architecture [22]. Former method consumes larger dynamic power due to increased switching activities and the later technique needs replication of architectures, resulting in larger area as well as higher power consumption, degrading energy efficiency of channel decoder in the physical layer. This article proposes to solve such design challenge in MAP decoder architecture where the achievable throughput can be varied without incorporating aforementioned methods. In brief, this work presents an MAP decoding algorithm and its reconfigurable architecture that operates in multiple-radix modes to deliver variable throughput at constant clock frequency. This enhances throughput and energy efficiency of the MAP decoder. Some of the key highlights of our contributions are as follows.

- 1) We propose algorithms for state-metrics and logarithmic likelihood ratio (LLR) computations for the MAP decoder that allows it to decode multiple hard bits in three different operating modes: radix-2, 4, and 8 generating 1, 2, and 3 decoded bits, respectively, in single clock cycle.
- 2) New microarchitectures for state-metric and LLR computation units are presented here. Furthermore, overall digital architecture of the MAP decoder has been proposed that can reconfigure its decoding operation in multiple radix fashion. Comprehensive power, timing, and throughput analyses in various CMOS technology nodes are also demonstrated in this article.
- 3) Performance analyses of the proposed MAP decoding algorithm that is operating at different modes have been carried out in additive-white Gaussian-noise (AWGN) channel environment.
- 4) The proposed multiradix MAP decoder has been fabricated in united-microelectronics-corporation (UMC) 130-nm-CMOS technology node. This ASIC chip occupies 2.325 mm² of die area and delivers a range of data-rates between 201 and 605 Mb/s. In addition, we have presented the hardware testing of our decoder prototype. Finally, the ASIC measured results are compared with the reported implementations in the literature.

II. PRELIMINARIES

Conventional physical layer of standard wireless communication system is shown in Fig. 1 that comprises three essential parts at both transmitter and receiver sides: digital baseband unit, radio frequency (RF) front end, and antenna. Our work focuses on the design of channel decoder which is key signal-processing block in the digital baseband unit of receiver. At the transmission side, datalink layer passes bitwise information to physical layer where it is source encoded to generate

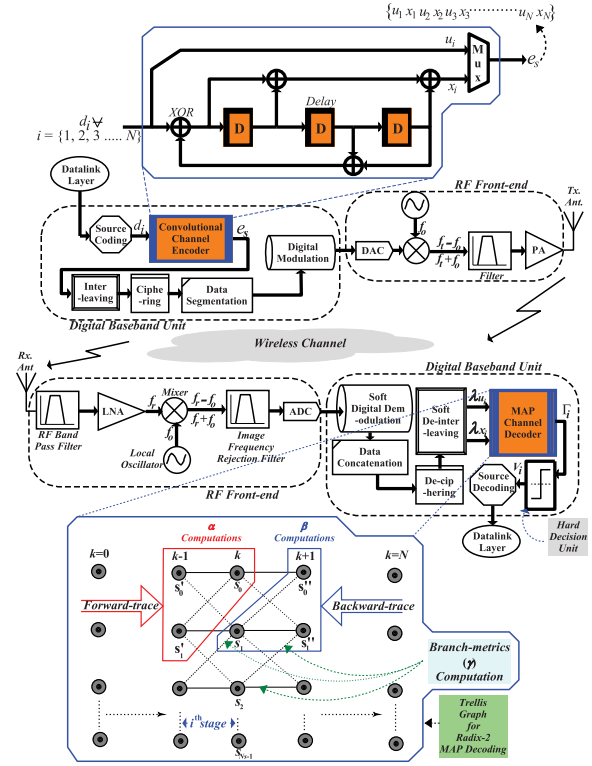


Fig. 1. Schematic of the physical layer for standard wireless-communication system where channel encoder and decoder are its inherent parts.

compressed bit-sequence of $d_i \forall i = \{1, 2, \dots, N\}$ where N represents the block length of convolutional channel code. Let us consider the transfer function $G(D)$ of convolutional channel encoder to be $[1, (1 + D + D^3)/(1 + D^2 + D^3)]$, which is compliant to 4G-LTE-Advanced standard [23], [24], with the constraint length of 4 and code-rate of 1/2, as shown in Fig. 1. This encoder generates systematic and parity bits u_i and $x_i \forall i = \{1, 2, \dots, N\}$ respectively. These bits are serialized as e_s and are passed through interleaver (to mitigate burst error), ciphering unit (for secure communication), data segmentation module (that segregates serial bits based on modulation scheme), and eventually modulated as well as digital-to-analog converted into an analog signal of baseband frequency f_i , as illustrated in Fig. 1. At the RF module, such baseband signal is upconverted (using mixer and local oscillator with the frequency f_0), power amplified; and finally transmitted via antenna. Received RF analog-signal at the output of receiver antenna is filtered, low-noise amplified, down converted to baseband signal of frequency f_r , before feeding to analog-to-digital converter (ADC), as shown in Fig. 1. These digitized output signals from ADC are soft-demodulated, concatenated, deciphered, and soft deinterleaved to generate LLRs for all the systematic and parity bits. Here, λ_{ui} and λ_{xi} represent received systematic and parity LLRs, respectively. These values are also known as *a priori* LLRs and are mathematically expressed as $\lambda_{ui} = \ln[P(u_i = 0|r)/P(u_i = 1|r)]$ and $\lambda_{xi} = \ln[P(x_i = 0|r)/P(x_i = 1|r)] \forall i = \{1, 2, \dots, N\}$ where r represents the received sequence of LLRs. Such *a priori* LLRs are fed to MAP channel decoder that generates *a posteriori* LLRs Γ_i

$\forall i = \{1, 2, \dots, N\}$ which are passed through hard decision unit to generate decoded bits $V_k \forall i = \{1, 2, \dots, N\}$, as shown in Fig. 1. Thereafter, V_i bits are source decoded and transferred to data-link layer at the receiver side.

The MAP channel-decoding is a trellis-graph based algorithm that has N stages and $N_s = 2^\varepsilon$ states where ε represents number of memory elements of convolutional encoder ($\varepsilon = 3$ in our work), as illustrated in Fig. 1. Here, each trellis stage contains all the possible encoder states that can be attained based on data input-bit d_i and initial encoder-state. Each state transition in i th trellis stage is associated with branch metric (γ_i) computation. Using these metrics, MAP decoding-algorithm foremost computes forward state metrics $\alpha_k(s_x) \forall x = \{0, 1, 2, \dots, N_s-1\}$ and $k = \{1, 2, \dots, N-1\}$, while forward tracing the trellis, as shown in Fig. 1. Here, the forward trace begins with the initialization of forward state metric at $k = 0$ as $\alpha_{k=0}(s_{x=0}) = \ln(1) = 0$ and $\alpha_{k=0}(s_{x \neq 0}) = \ln(0) = -\infty$. Thereafter, the backward trace begins for computing backward state metrics $\beta_k(s_x) \forall x = \{0, 1, 2, \dots, N_s-1\}$ and $k = \{N-1, N-2, \dots, 1\}$. Now assuming i th stage in the trellis graph from Fig. 1, the expressions for forward and backward state-metrics for S_0 and S_1 states, respectively, are computed as

$$\alpha_k(S_0) = \max\{\alpha_{k-1}(S'_0) + \gamma_i(S'_0, S_0), \alpha_{k-1}(S'_1) + \gamma_i(S'_1, S_0)\} \quad (1)$$

$$\beta_k(S_1) = \max\{\beta_{k+1}(S''_0) + \gamma_i(S''_0, S_1), \beta_{k+1}(S''_1) + \gamma_i(S''_1, S_1)\}. \quad (2)$$

Similarly, branch metrics for the state transition $S_x \rightarrow S_y$ in i th trellis stage are computed as $\gamma_i(S_x, S_y) = l_c/2 \times \{u_i \times \lambda u_i + x_i \times \lambda x_i\}$ where l_c is a channel reliability factor [25]. Here, $\{u_i, x_i\}$ are the generated systematic and parity bits, respectively, when encoder state changes from $S_x \rightarrow S_y$. Eventually, *a posteriori* LLRs $\Gamma_i \forall i = \{1, 2, \dots, N\}$ are determined after the computation of state and branch metrics for all trellis stages. Assuming δ as all possible trellis transitions, where $s^{st}(\delta)$ and $s^{en}(\delta)$ correspond to start and end states, Γ_i value is computed as

$$\Gamma_i = \max_{\delta: (s', s) \Rightarrow u_i=1} \{f(\delta)\} - \max_{\delta: (s', s) \Rightarrow u_i=0} \{f(\delta)\} \quad (3)$$

where the function $f(\delta)$ is expressed as

$$f(\delta) = \alpha_{k-1}\{s^{st}(\delta)\} + \gamma_k(\delta) + \beta_k\{s^{en}(\delta)\}. \quad (4)$$

Additionally, $\delta: (s', s) \Rightarrow u_i = 0/1$ indicates set of all trellis transitions when the information bit is “0” or “1,” respectively. It is to be noted that aforementioned channel decoding is based on the max-log MAP radix-2 algorithm which generates single LLR in each clock cycle [22].

III. PROPOSED ALGORITHMS AND ARCHITECTURES

A. Multiple-Radix State-Metric Computations

Proposed MAP-decoding algorithm and its architecture have been designed using the sliding-window (SW) technique for N block size and M SW size [25]. As discussed earlier in Fig. 1, soft demodulator generates $2 \times N$ *a priori* LLRs (N values each for λu_i and λx_i) and are fed as inputs to the MAP

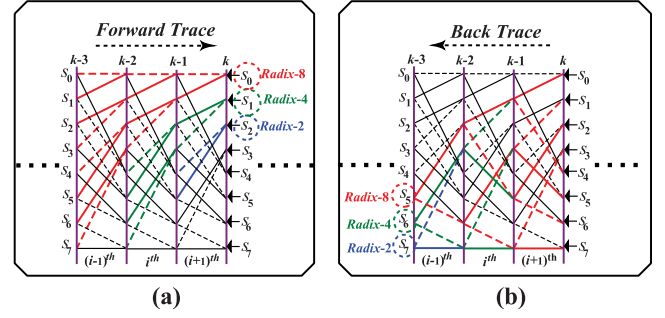


Fig. 2. Trellis graph of radix-2/4/8 (a) forward and (b) backward state-metrics MRC for multiradix MAP decoding algorithm.

decoder. For the trellis graph used in MAP decoder, there are $2 \times N_s$ state transitions in each trellis stage and every transition is associated with a branch metric computation. Rather computing $2 \times N_s$ branch metrics in a given trellis stage, we compute only $\log_2(2 \times N_s)$ parent branch metrics (PBMs) and use them in repetitive fashion to cover all state transitions [22]. We propose to reconfigure the computations of state metrics $-\alpha_k(S_x)$ & $\beta_k(S_x)$ in radix-2/4/8 mode and this process is termed as multiple radix computation (MRC). As discussed earlier, radix-2 MAP decoding processes single trellis stage at a given time; on the other side, radix-4 and radix-8 MAP decoding process two and three trellis stages, respectively, in a time instance. Hence, such MRC model of MAP decoding can deliver 1/2/3 *a posteriori* LLRs (Γ_i) at a given time instance. For clarity, let us assume an arbitrary three stages of trellis graph which has $N_s = 8$ states and state-transitions based on G(D), as discussed in Section II. Let s_0, s_1 and s_2 represent three different states for radix-8, 4 and 2 forward state-metric computations, respectively, at k th column of $(i+1)$ th trellis stage shown in Fig. 2(a). Therefore, forward state-metric-computation for radix-8 MRC at state s_0 is calculated as [referring Fig. 2(a)]

$$\alpha_k(s_0) = \max\{\alpha_{k-1}(s_0) + \gamma_{i+1}(s_0, s_0), \alpha_{k-1}(s_1) + \gamma_{i+1}(s_1, s_0)\} \quad (5)$$

where

$$\alpha_{k-1}(s_0) = \max\{\alpha_{k-2}(s_0) + \gamma_i(s_0, s_0), \alpha_{k-2}(s_1) + \gamma_i(s_1, s_0)\} \quad (6)$$

$$\alpha_{k-1}(s_1) = \max\{\alpha_{k-2}(s_2) + \gamma_i(s_2, s_1), \alpha_{k-2}(s_3) + \gamma_i(s_3, s_1)\}. \quad (7)$$

Furthermore,

$$\begin{aligned} \alpha_{k-2}(s_0) &= \max\{\alpha_{k-3}(s_0) + \gamma_{i-1}(s_0, s_0), \alpha_{k-3}(s_1) + \gamma_{i-1}(s_1, s_0)\} \end{aligned} \quad (8)$$

$$\begin{aligned} \alpha_{k-2}(s_1) &= \max\{\alpha_{k-3}(s_2) + \gamma_{i-1}(s_2, s_1), \alpha_{k-3}(s_3) + \gamma_{i-1}(s_3, s_1)\} \end{aligned} \quad (9)$$

$$\begin{aligned} \alpha_{k-2}(s_2) &= \max\{\alpha_{k-3}(s_4) + \gamma_{i-1}(s_4, s_2), \alpha_{k-3}(s_5) + \gamma_{i-1}(s_5, s_2)\} \end{aligned} \quad (10)$$

$$\begin{aligned} \alpha_{k-2}(s_3) &= \max\{\alpha_{k-3}(s_6) + \gamma_{i-1}(s_6, s_3), \alpha_{k-3}(s_7) + \gamma_{i-1}(s_7, s_3)\}. \end{aligned} \quad (11)$$

On the other hand, radix-4 MRC of forward state-metric for the state s_1 is carried out as

$$\alpha_k(s_1) = \max\{\alpha_{k-1}(s_2) + \gamma_{i+1}(s_2, s_1), \alpha_{k-1}(s_3) + \gamma_{i+1}(s_3, s_1)\} \quad (12)$$

where

$$\alpha_{k-1}(s_2) = \max\{\alpha_{k-2}(s_4) + \gamma_i(s_4, s_2), \alpha_{k-2}(s_5) + \gamma_i(s_5, s_2)\} \quad (13)$$

$$\alpha_{k-1}(s_3) = \max\{\alpha_{k-2}(s_6) + \gamma_i(s_6, s_3), \alpha_{k-2}(s_7) + \gamma_i(s_7, s_3)\}. \quad (14)$$

Eventually, the radix-2 MRC of forward-state metric for s_2 is obtained as

$$\alpha_k(s_2) = \max\{\alpha_{k-1}(s_4) + \gamma_{i+1}(s_4, s_2), \alpha_{k-1}(s_5) + \gamma_{i+1}(s_5, s_2)\}. \quad (15)$$

The MRC of radix-2, 4, and 8 backward state metrics are performed in similar fashion while backward tracing the trellis graph. Fig. 2(b) shows the back trace of same three stages where s_5 , s_6 , and s_7 are states of $(i-1)$ th trellis stage that represent radix-8, 4, and 2 MRC, respectively. As discussed above, these backward state metrics are computed as

$$\beta_k(s_5) = \max\{\beta_{k-2}(s_2) + \gamma_{i-1}(s_2, s_5), \beta_{k-2}(s_6) + \gamma_{i-1}(s_6, s_5)\} \quad (16)$$

for radix-8 MRC

$$\beta_k(s_6) = \max\{\beta_{k-2}(s_3) + \gamma_{i-1}(s_3, s_6), \beta_{k-2}(s_7) + \gamma_{i-1}(s_7, s_6)\} \quad (17)$$

for radix-4 MRC and finally

$$\beta_k(s_7) = \max\{\beta_{k-2}(s_3) + \gamma_{i-1}(s_3, s_7), \beta_{k-2}(s_7) + \gamma_{i-1}(s_7, s_7)\} \quad (18)$$

for radix-2 MRC.

B. Microarchitecture of MRC State-Metric Computation-Unit (MSMCU)

The proposed MSMCU of MAP decoder computes both forward and backward state metrics for each trellis state. Thereby, stack of N_s such MSMCUs are required for computing all the state metrics of each trellis stage. First, we propose a flexible MSMCU architecture that computes three possible forward/backward state metrics for radix-2/4/8 MAP decoding, respectively, as shown in Fig. 3. For the ease of understanding, we have only considered the forward state-metric computation for realizing $\alpha_k(s_0)$, $\alpha_k(s_1)$, and $\alpha_k(s_2)$ based on their expressions from (5), (12), and (15), respectively. Here, the key challenge is to reliably route branch and state metrics for different operating modes. As we scale-up the radix computation from radix-2 to radix-8, traversing of trellis becomes deeper for the state metric computation of present state. At the same time, the number of branch as well as state metrics of previous states—those are required for state metric calculation—gradually increases.

Rad2-SMCU of MSMCU architecture computes the state metric for radix-2 operating mode from (15), as shown

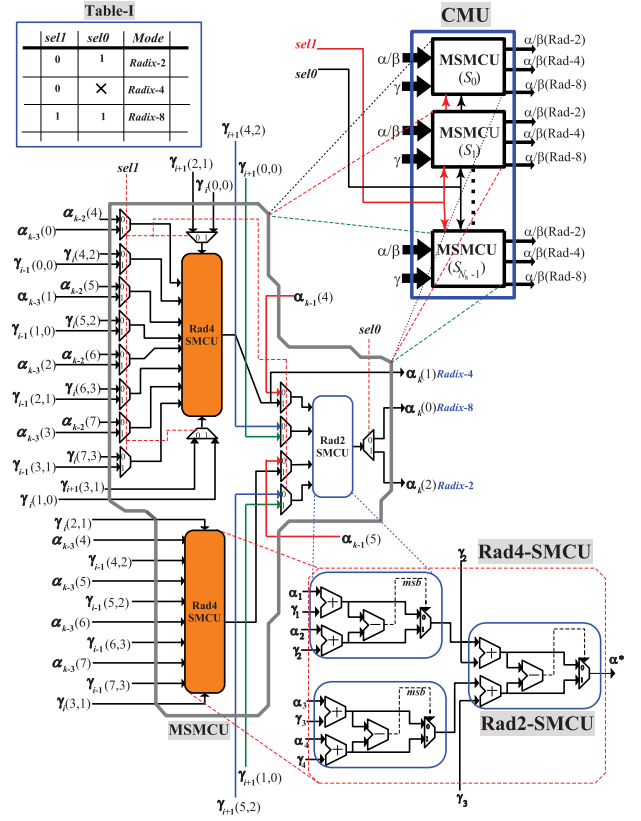


Fig. 3. Proposed digital architecture of reconfigurable MSMCU and CMU for radix-2/4/8 operating modes. Here, the trellis states are represented by numbers; for example: states $S_0, S_1, \dots, S_7$ are denoted as $0, 1, \dots, 7$.

in Fig. 3. Three such units are aggregated to construct Rad4-SMCU block for the calculation of radix-4 state metric using (12)–(14). These Rad4-SMCU blocks are integrated with the multiplexer/demultiplexer network to build reconfigurable MSMCU which computes radix-8/4/2 state metrics for a given state depending on the values of select lines $sel0$ and $sel1$, as shown in Table-I of Fig. 3. For example, when $sel1$ is high then all the state metrics of $(k-3)$ th column and branch metrics of $(i-1)$ th stage, for radix-8 computation, are passed to Rad4-SMCU blocks through first stage of multiplexers. Subsequently, next multiplexer-stage releases two computed-radix-4 state-metrics which are then fed to Rad2-SMCU along with respective branch metrics. Eventually, if $sel0$ is high then de-multiplexer releases radix-8 state metric $\alpha_k(S_0)$, as shown in Fig. 3. Similarly, this architecture can be used for radix-2/4 state-metric computation by changing the values of select lines. Alternately, the suggested MSMCU can perform MRC of backward state metrics while reverse tracing the trellis graph. Finally, we aggregate N_s MSMCUs to design combined-MSMCU (CMU) shown in Fig. 3. Such CMU architecture generates either of the radix-2/4/8 forward or backward state-metrics based on the reconfigurable selection inputs (Fig. 3 and $sel0$). Integration of CMUs in our MAP decoder architecture has been discussed in Section III-D.

C. Mathematical Formulations and Flexible Architecture of A Posteriori LLR Computation

Computation of *a posteriori* LLR(s) supporting multiple radix operation in the proposed MAP decoder is crucial

and complex operation. Our flexible decoder processes the information at radix-2, 4, and 8 modes to deliver single, double, and triple LLR(s), respectively, in a given computation cycle. In general, the transition probability (Π) of any arbitrary state transition $s_p \rightarrow s_q$ for radix-2 MAP decoding (referring $(i+1)$ th trellis stage from Fig. 2) is computed as

$$\Pi_{p,q} = \alpha_{k-1}(p) + \gamma_{i+1}(s_p, s_q) + \beta_k(q) \quad \forall \{p, q\} \in N_s. \quad (19)$$

Using these transition probabilities, radix-2 *a posteriori* LLR of $(i+1)$ th stage from Fig. 2 is computed as

$$\Gamma_{i+1} = \max_{\delta u_j=1} \{\Pi_{p,q}\} - \max_{\delta u_j=0} \{\Pi_{p,q}\} = \Pi_{p,q}^1 - \Pi_{p,q}^0 \quad (20)$$

where δ : $u_j = 1/0$ represents the set of all state transitions when u_j equals 0 or 1. For radix-4 LLR computation, two radix-2 trellis-stages are processed together, as illustrated in Fig. 2. If $s_{q'}$ represents an intermediate state between $s_p \rightarrow s_q$ state-change then its transition probability is given by

$$\Pi_{p,q':q} = \alpha_{k-3}(p) + \gamma_{i-1}(s_p, s_{q'}) + \gamma_i(s_{q'}, s_q) + \beta_{k-1}(q). \quad (21)$$

There are four state transitions from previous stage toward each of the sink states in radix-4 trellis, as presented in trellis graph from Fig. 2. Thereby, four transition probabilities for each of such states are represented as $\Pi_{p,q':q}^{00}$, $\Pi_{p,q':q}^{10}$, $\Pi_{p,q':q}^{11}$ and $\Pi_{p,q':q}^{01}$. Thereby, *a posteriori* LLRs for $(i-1)$ th and i th trellis stages which are computed in single computation cycle are

$$\Gamma_{i-1} = \max_{\delta u_j=1} \{\Pi_{p,q':q}^{10}, \Pi_{p,q':q}^{11}\} - \max_{\delta u_j=0} \{\Pi_{p,q':q}^{00}, \Pi_{p,q':q}^{01}\} \quad (22)$$

$$\Gamma_i = \max_{\delta u_j=1} \{\Pi_{p,q':q}^{11}, \Pi_{p,q':q}^{01}\} - \max_{\delta u_j=0} \{\Pi_{p,q':q}^{00}, \Pi_{p,q':q}^{10}\} \quad (23)$$

respectively. Thus, the radix-4 MAP decoding generates two *a posteriori* LLRs—corresponding to two decoded bits—in each clock cycle. On the other side, there are eight state-transitions toward each of the sink states in radix-8 trellis stage, as shown in Fig. 2, where each of its corresponding transition probability is computed as

$$\Pi_{p:r:s,q} = \alpha_{k-3}(p) + \gamma_{i-1}(s_p, s_r) + \gamma_i(s_r, s_s) + \gamma_{i+1}(s_s, s_q) + \beta_k(q) \quad (24)$$

where s_r and s_s are intermediate states at $(k-2)$ and $(k-1)$ columns of $(i-1)$ th and i th trellis stages, respectively. Therefore, three *a posteriori* LLRs computed in each computation cycle of radix-8 decoding are formulated as

$$\Gamma_{i-1} = \max_{\delta u_j=1} \{\Pi_{p:r:s,q}^{100}, \Pi_{p:r:s,q}^{110}, \Pi_{p:r:s,q}^{111}, \Pi_{p:r:s,q}^{101}\} - \max_{\delta u_j=0} \{\Pi_{p:r:s,q}^{000}, \Pi_{p:r:s,q}^{010}, \Pi_{p:r:s,q}^{011}, \Pi_{p:r:s,q}^{001}\} \quad (25)$$

$$\Gamma_i = \max_{\delta u_j=1} \{\Pi_{p:r:s,q}^{110}, \Pi_{p:r:s,q}^{010}, \Pi_{p:r:s,q}^{011}, \Pi_{p:r:s,q}^{111}\} - \max_{\delta u_j=0} \{\Pi_{p:r:s,q}^{000}, \Pi_{p:r:s,q}^{100}, \Pi_{p:r:s,q}^{101}, \Pi_{p:r:s,q}^{001}\} \quad (26)$$

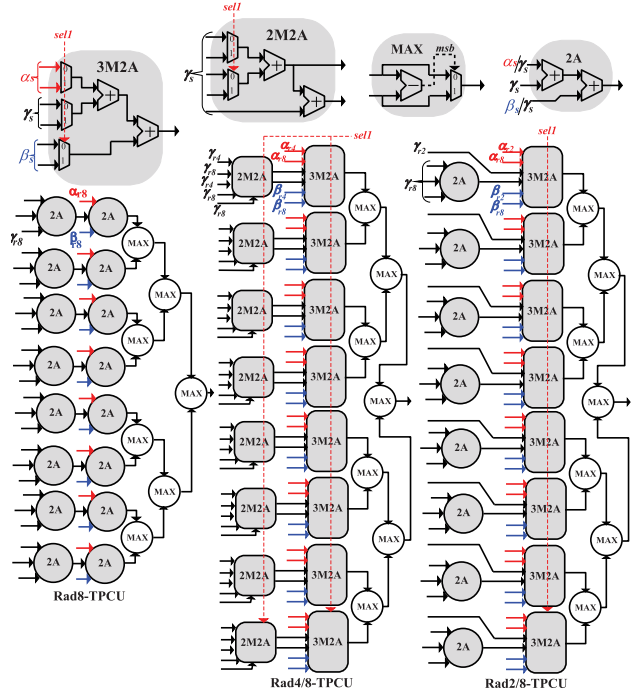


Fig. 4. Proposed reconfigurable digital-architectures of submodules used for computing transition probabilities in MLCU.

$$\Gamma_{i+1} = \max_{\delta u_j=1} \{\Pi_{p:r:s,q}^{011}, \Pi_{p:r:s,q}^{111}, \Pi_{p:r:s,q}^{101}, \Pi_{p:r:s,q}^{001}\} - \max_{\delta u_j=0} \{\Pi_{p:r:s,q}^{000}, \Pi_{p:r:s,q}^{100}, \Pi_{p:r:s,q}^{110}, \Pi_{p:r:s,q}^{010}\}. \quad (27)$$

The transition probabilities $\Pi_{p,q}$, $\Pi_{p,q':q}$, and $\Pi_{p:r:s,q}$ from (19), (21), and (24), respectively, indicate that the number of branch metrics for their computations varies for different operating modes of multiradix MAP decoding. Prior delving into the design of multiradix LLR-computation unit (MLCU), we present the architectures of three integral submodules of MLCU: radix-8 transition-probability computation-unit (Rad8-TPCU), flexible-radix-4/8 transition-probability computation-unit (Rad4/8-TPCU), and flexible-radix-2/8 transition-probability computation-unit (Rad2/8-TPCU). These architectures are shown in Fig. 4 where γ_{r8} , γ_{r4} , and γ_{r2} denote branch metrics for radix-8, 4, and 2 operating modes, respectively, and the corresponding forward/backward state metrics are denoted by α_{r8}/β_{r8} , α_{r4}/β_{r4} , and α_{r2}/β_{r2} . Fig. 4 also presents the architectures for 3M2A, 2M2A, 2A and MAX internal submodules. Suggested architectures of Rad4/8-TPCU and Rad2/8-TPCU can be reconfigured using *sell* signal (the same select signal that is used in CMU, as shown in Fig. 3) to compute transition probabilities of radix-4/8 (i.e., $\Pi_{p,q':q}$ or $\Pi_{p:r:s,q}$) and radix-2/8 ($\Pi_{p,q}$ or $\Pi_{p:r:s,q}$) operating modes, respectively. Similarly, Rad8-TPCU computes transition probabilities for only radix-8 mode (i.e., $\Pi_{p:r:s,q}$). The network of MAX comparators is interconnected with 2A, 2M2A, and 3M2A units to construct the architectures for Rad4/8-TPCU, Rad2/8-TPCU and Rad8-TPCU, as shown in Fig. 4. These modules along with subtractors as well as MAX comparators are aggregated to construct MLCU architecture, as shown in Fig. 5. It is built

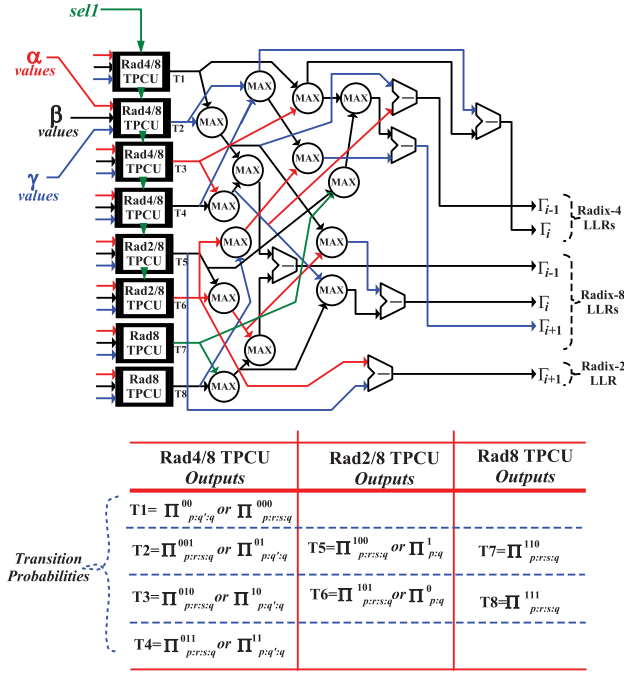


Fig. 5. Proposed digital-architecture of MLCU for generating *a posteriori* LLRs for radix-2/4/8 operating mode.

using four Rad4/8-TPCUs, two Rad2/8-TPCUs, two Rad8-TPCUs and their outputs are denoted as $\{T1, T2, T3, \dots, T8\}$ representing the transition probabilities, as tabulated in Fig. 5. These values are processed based on (20), (22), (23), and (25)–(27) for computing *a posteriori* LLRs for radix-2, 4, and 8 operating modes, respectively, from the suggested MLCU architecture shown in Fig. 5.

D. Proposed MAP Decoder-Architecture for Multiradix Operation

SW technique-based MAP decoding algorithm is extensively used for its real-world implementations [25]. In this technique, all N stages of the trellis graph are segregated into multiple SWs where each SW is a collection of M trellis stages (i.e., there are N/M SWs). The proposed multiradix MAP-decoder has been designed based on this technique, as shown in Fig. 6. In this architecture, received *a priori* LLRs λ_{ui} and λ_{xi} , with the bit-width of n_λ , are first processed by the branch-metrics computation-unit (BMCU) that generates $\varphi = \log_2(2 \times N_s)$ PBMs of n_γ bit-width. These metrics are concatenated to form ψ_{con} value which has a bit width of $m_\psi = \varphi \times n_\gamma$, as shown in Fig. 6. Furthermore, the operation of proposed MAP decoder-architecture is presented as follows, using the timing chart from Fig. 7. Here, each time slot represents the processing time of single SW.

- 1) *SW1*: In this time slot, PBMs of M trellis stages of *SW1* are computed by BMCU and are stored in *Mem-1* memory of size $M \times m_\psi$.
- 2) *SW2*: The PBMs of *SW2* are generated by BMCU and stored in *Mem-2* memory.
- 3) *SW3*: In this time slot, three operations are carried out simultaneously.

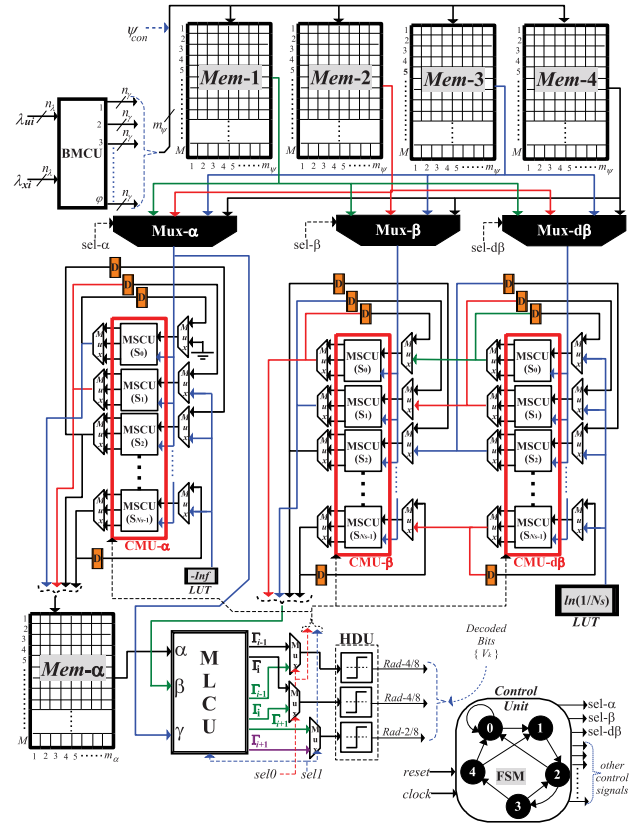


Fig. 6. Proposed reconfigurable architecture of MAP decoder for multiple radix-2/4/8 operation.

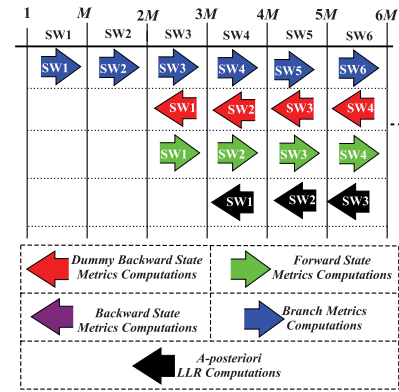


Fig. 7. Timing chart illustrating the SW flow of MAP decoding algorithm for the proposed decoder.

- a) PBMs for *SW3* are computed and stored in *Mem-3*.
- b) The CMU has been used in feedback configuration mode (referred as CMU-dβ), as shown in Fig. 6. Initial value $\ln(1/N_s)$ of backward state metrics for the last trellis-stage of *SW2*, that initiates the dummy backward state-metrics computations, has been stored in *LUT* and are fed to CMU-dβ via multiplexer. Here, PBMs are read from *Mem-2* and are fed to CMU-dβ via Mux-dβ that computes dummy backward state-metrics for M th stage of *SW1*.
- c) Similarly, CMU-α are also fed with initial values of zero and $-\infty$ (i.e., largest negative value),

as shown in Fig. 6. At the same time, PBMs of SW1 from *Mem-1* are also provided to CMU- α via Mux- α to compute the forward state metrics of all stages in SW1 and are stored in *Mem- α* , as shown in Fig. 6. Timing chart presented in Fig. 7 can be referred for better understanding of these SW-based MAP decoding flow.

- 4) *SW4*: As shown in Fig. 7, we again compute the PBMs for all the stages of SW4 and store them in *Mem-4*. Here, *Mem-3* is read to provide PBMs of SW2, via Mux-d β , to CMU-d β for computing dummy backward state-metrics of SW2. Here, PBMs from *Mem-2* are also read to feed these values to CMU- α , via Mux- α , for computing the forward state metrics of SW2 and are stored in *Mem- α* . Simultaneously, *Mem- α* (it is a dual-port memory) is read to extract the forward state metrics of SW1 and are fed to MLCU, as shown in Fig. 6. Using the dummy backward state metrics and the PBMs of SW1 read from *Mem-1* (routed via Mux- β), the actual backward state metrics for SW1 are computed using CMU- β and are fed to MLCU. Thus, forward and backward state metrics of SW1 are fed to MLCU along with PBMs of SW1 via Mux- α and these information suffice MLCU to start computing the *a posteriori* LLRs of M trellis stages, as shown in Fig. 6. In addition, *sel1* and *sel0* signals can be used for selecting the radix-2/4/8 mode of operation, as discussed earlier in Fig. 3, and these select signals are also used at the multiplexer network of MLCU.
- 5) *SW5*: Here, the PBMs for SW5 are computed and stored in *Mem-1*. Similarly, PBMs from *Mem-4* are read and are fed to CMU-d β that computes dummy backward state-metrics for SW3. At the same time, CMU- α calculates forward state metrics for SW3 and stores in *Mem- α* . Forward state metrics of SW2 are also read from *Mem- α* and are fed to MLCU. Similar to SW5 time slot, M *a posteriori* LLRs for SW2 are computed by MLCU. In subsequent SWs, a similar process is repeated and continues till all the *a posteriori* LLRs are computed, as illustrated in Fig. 7.

Finally, *a posteriori* LLRs from MLCU are fed to hard decision unit (HDU) that generates decoded bits $V_k \forall k = \{1, 2, \dots, N\}$, as shown in Fig. 6. In addition, the proposed MAP decoder consists of Mealy finite-state-machine-based control unit that outputs required control signals (such as *sel- α* , *sel- β* , and *sel-d β*) for various modules in our MAP decoder, as shown in Fig. 6.

E. Timing, Power and Throughput Analyses

The proposed multiradix MAP decoder-architecture has been designed for $N = 6144$ bits and $M = 24$ trellis stages (i.e., $\approx 5\text{-}6 \times$ constraint-length of convolutional encoder [25]). From the timing chart of SW-based MAP decoding shown in Fig. 7, *a posteriori* LLRs are generated by MLCU after $3 \times M$ clock cycles which is the latency Λ of our decoder and is expressed as $\Lambda = 3 \times M$ clock cycles. For the multiradix operation, ρ represents the number of decoded bits that

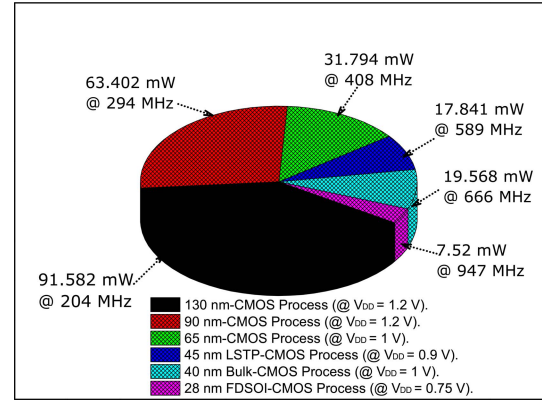


Fig. 8. Total power consumptions of the proposed multiradix MAP decoder when synthesized and postlayout simulated in various CMOS technology nodes.

can be delivered by our decoder in single clock cycle and thereby, radix-2/4/8 modes can be expressed as $\text{radix-}2^\rho \forall \rho = \{1/2/3\}$. Therefore, if $F_{\max}$ denotes maximum clock-frequency then the decoding throughput Θ_τ is computed as

$$\Theta_\tau = \left(\frac{\rho \times N \times F_{\max}}{N + \Lambda} \right). \quad (28)$$

Above expression of Θ_τ indicates unique feature of the proposed multiradix MAP-decoder where the decoding throughput can be varied by operating the decoder in different modes (using the *sel1* and *sel0* input signals of our decoder architecture from Fig. 6) without changing the maximum operating clock frequency $F_{\max}$ and maintaining constant power dissipation (specifically the dynamic power).

The suggested MAP-decoder architecture is register-transfer-level (RTL) coded, synthesized, static-timing analyzed and post-layout simulated using standard electronic-design-automation (EDA) tools in six different technology nodes: 130-nm CMOS, 90-nm CMOS, 65-nm CMOS, 45-nm low-standby-power (LSTP)-CMOS, 40-nm Bulk-CMOS, and 28-nm fully depleted silicon-on-insulator (FDSOI)-CMOS technology nodes. Fig. 8 shows the total power consumed by our design at different technology nodes with the supply voltages ranging from 0.75 to 1.2 V. In addition, it presents the maximum clock frequencies obtained after the static timing analysis of our decoder architecture that ranges between 204 and 947 MHz. It is to be noted that the total power consumptions shown in Fig. 8 are obtained, at maximum clock frequencies, for processing ten blocks of *a priori* LLRs where each block comprises 6144 LLRs. It illustrates that the maximum power of 91.6 mW at 204 MHz is consumed by our decoder in 130-nm-CMOS process. Furthermore, we have determined the decoding throughputs achieved by the suggested decoder architecture, based on (28), for various technology nodes, as shown in Fig. 9. Here, three different decoding-throughputs are computed for radix-2 ($\rho = 1$), 4 ($\rho = 2$), and 8 ($\rho = 3$) operating modes at each technology node, operating at corresponding maximum clock-frequencies presented in Fig. 8. It is obvious from Fig. 9 that the maximum range of throughputs (0.94–2.8 Gb/s) have been achieved by

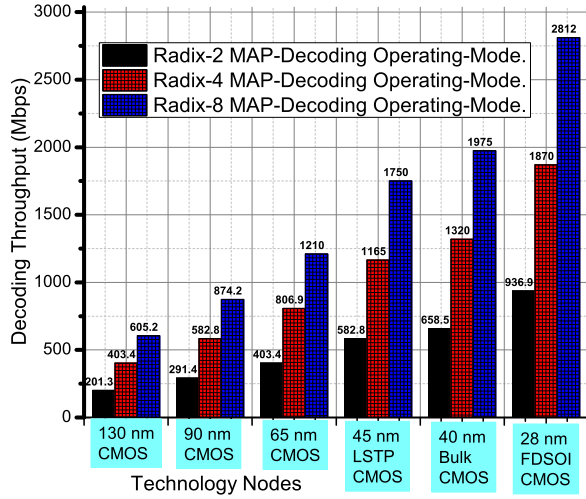


Fig. 9. Analyses of achievable throughputs obtained by the proposed MAP decoder, while operating in radix-2, 4, and 8 mode, at different CMOS technology nodes.

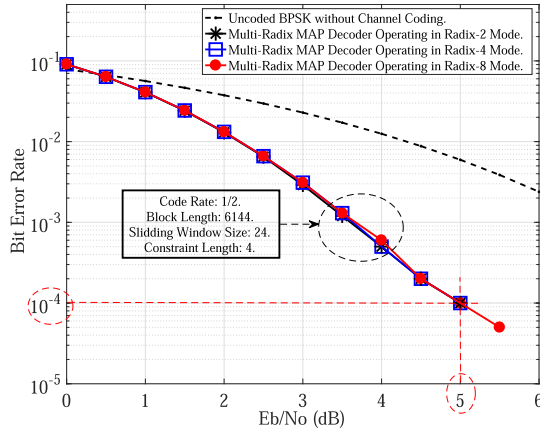


Fig. 10. BER performance analysis of the proposed MAP decoding algorithm operating at radix-2, 4, and 8 modes.

our design, when synthesized and post-layout simulated in 28-nm FDSOI-CMOS technology node.

IV. PERFORMANCE ANALYSIS, ASIC IMPLEMENTATION, HARDWARE TESTING, AND COMPARISONS

A. Performance Analyses

Extensive high-level simulations have been carried out for analyzing bit-error-rate (BER) performance of the proposed MAP decoding algorithm in AWGN channel-environment based communication system, as presented in Fig. 1. Here, the information bits are modulated (at transmitter side) and soft-demodulated (at receiver side) using binary phase shift-keying (BPSK) technique. Here, Fig. 10 shows the BER plots for multiradix-2, 4, and 8 modes of MAP decoding algorithm while varying the channel SNR or E_b/N_0 from 0 to 5.5 dB. During these high-level Monte Carlo simulations, we have transmitted $\approx 10^8$ information bits and are MAP decoded at every E_b/N_0 values between 0–5.5 dB with an interval of 0.5 dB. Simulated BER plots of MAP decoding algorithm operating in radix-2, 4, and 8 modes show no performance loss

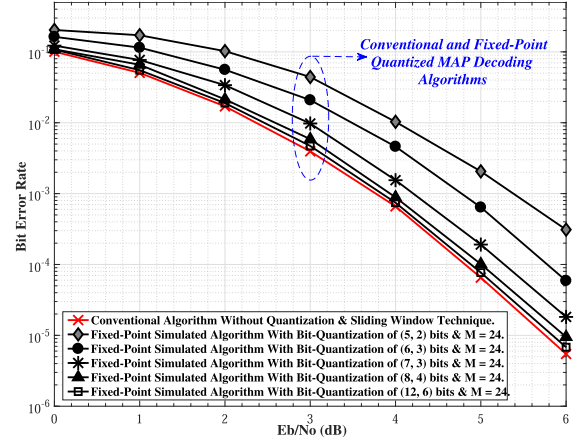


Fig. 11. Comparative BER performance analysis of the conventional and fixed-point versions of proposed MAP decoding algorithms, operating at radix-2 mode.

among them, as shown in Fig. 10. It can be marked that the proposed MAP decoding algorithm at all three modes delivers a BER of 10^{-4} at an E_b/N_0 of 5 dB.

With the aforementioned specifications and channel environment, this work further presents fixed-point simulation based BER performance analysis of the proposed MAP decoding algorithm, with five different values of n_λ , and their comparison with the conventional algorithm. The BER plots obtained from such analysis are shown in Fig. 11 where both conventional (with full-precision floating point representation) and proposed fixed-point algorithms are operated in radix-2 mode. Here, the conventional MAP decoding-algorithm has been simulated without incorporating SW technique (i.e., the trellis graph is forward traced till N th stage and subsequently, traced back from this stage). As shown in Fig. 11, the proposed MAP decoding algorithms with $n_\lambda = (12, 6)$ bit and $n_\lambda = (8, 4)$ bit deliver equivalent performances to conventional algorithm with the coding losses of 0.06 and 0.18 dB, respectively, at a BER of 10^{-4} . On the other side, algorithms with the quantization of (7, 3), (6, 3), and (5, 2) bit deliver degraded performance of 0.43, 0.97, and 1.57 dB at a BER of 10^{-3} , respectively, with respect to conventional algorithm. Therefore, this work adopted a bit-quantization of $n_\lambda = (8, 4)$ bit—where 8 bit denotes total bit-width of LLR and 4 bit represent its fractional precision—for the proposed MAP-decoder architecture, as it achieved adequate BER performance with minor coding loss.

B. ASIC Implementation

The proposed MAP decoder-architecture has been functionally verified and synthesized using the standard cell libraries of UMC 130-nm-CMOS process (FSC0H_D logic HS-FSG process). Subsequently, its gate-level netlist is imported for the physical design using standard EDA tool. The digital layout of our design is exhaustively design-rule-check (DRC) verified (using the DRC-rule files from UMC) and post-layout simulated before generating its graphic-data-stream (GDS)-II file. Finally, this design has been fabricated, in UMC 130-nm-CMOS technology node, into an ASIC chip of die

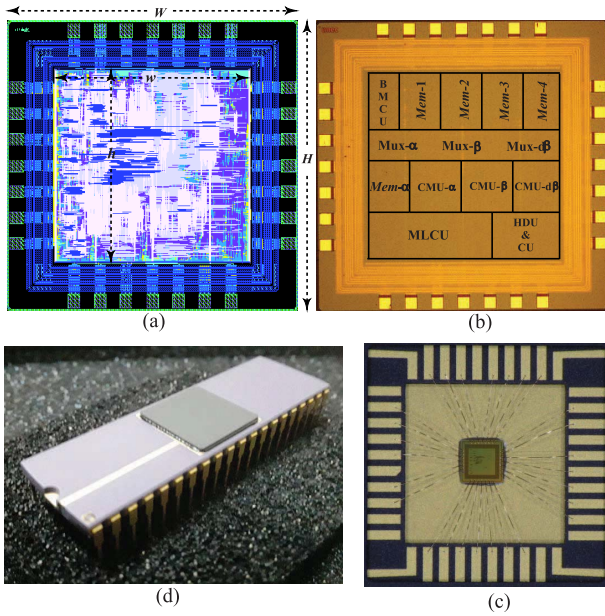


Fig. 12. (a) Chip layout with core dimension $(h \times w) = (1134 \mu\text{m} \times 1132.8 \mu\text{m})$ and die dimension $(H \times W) = (1525 \mu\text{m} \times 1525 \mu\text{m})$. (b) Chip micrograph. (c) Microscopic image of packaged chip. (d) Final packaged ASIC-prototype of the proposed MAP decoder fabricated in UMC 130-nm-CMOS technology node.

area 2.325 mm^2 . The CMOS layout, die micrograph, packaged die-micrograph, and final CD_DIL40 package version of our ASIC chip are shown in Fig. 12(a)–(d). It consists of 28 pads: eight pads for 8-bit LLR λ_{ui} , eight pads for another 8-bit LLR λ_{xi} , two pads for *sel1* and *sel0* signals for reconfiguring the radix-2/4/8 operating modes, one pad for reset signal, one pad for master clock, three pads for decoded output bits for different radix operations, three pads for core and (input and output) pad supplies, and two pads for their grounds. As shown in Fig. 12, this ASIC chip has a core area of 1.284 mm^2 and can operate with a maximum clock frequency of 204 MHz. As discussed earlier, the fabricated chip is capable of decoding convolutional code of block length 6144 bits with a code rate of $1/2$ and an SW size of 24. Therefore, it delivers data-rates of 201, 403, and 605 Mb/s while operating in radix-2, 4, and 8 modes, respectively, based on (28). These throughputs are achievable at a single clock frequency of 204 MHz consuming a total power of 92 mW.

Due to reconfigurable architecture of the proposed MAP decoder, the higher radix operation that enhances the decoder-throughput can be achieved by changing the values of *sel1* and *sel0* signals, as discussed earlier in Section III-B. Necessity of increasing the clock frequency to attain such higher throughput can be deferred in our decoder architecture. Therefore, it consumes lesser power while operating at lower clock frequency and delivers higher throughput. In addition, this work also incorporated automated clock-gating technique in our ASIC design flow to further minimize the power consumption. As shown in Fig. 6, various submodules of suggested MAP decoder like BMCU, *Mem-1/2/3/4*, *Mem- α* , *CMU- $\alpha/\beta/d\beta$* , MLCU and HDU, need not operate simultaneously. Hence, these modules are graphically scheduled using the sequencing

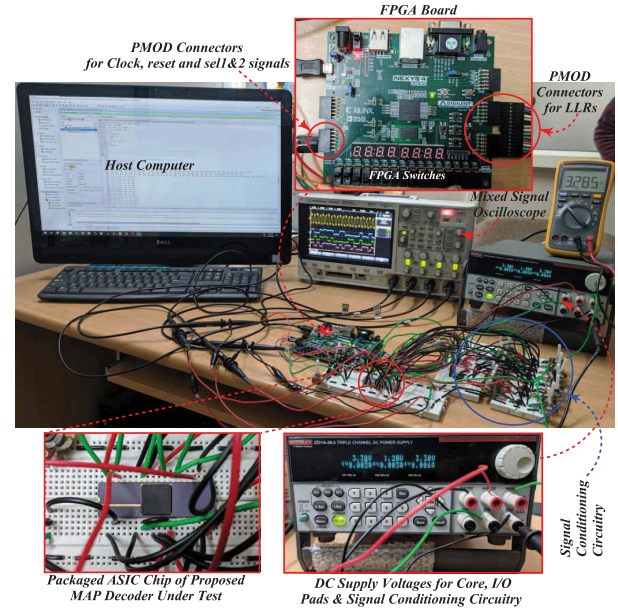


Fig. 13. Hardware test-setup for testing the fabricated ASIC chip of proposed multiradix MAP decoder.

graph and the clock signals are systematically gated to avoid unnecessary switching activities. Aforementioned technique further enabled our decoder architecture to consume lesser power.

C. Chip Testing

In order to generate input LLRs (λ_{ui} and λ_{xi}) for testing the decoder chip, communication system shown in Fig. 1 has been implemented using MATLAB Simulink tool in the host computer. At a given SNR, $\approx 10^6$ received λ_{ui} and λ_{xi} input-LLR values are saved. These LLRs are bit-quantized into 8-bit fixed-point numbers and are stored in two block random access-memories (BRAMs) of 8 MB each from Nexsys-4 DDRTM field-programmable gate array (FPGA) board (with Xilinx ARTIX-7 XC7A100T part) using the Vivado tool, as shown in Fig. 13. It shows the complete hardware test-setup where the FPGA board (using PMOD connectors) has been interfaced with our chip via signal conditioning circuitry. The FPGA internal-clock that operates at 200 MHz is tapped out via PMOD pin and is fed as master input-clock signal to our chip. Fig. 13 shows that FPGA switches have been used for feeding reset as well as *sel1* and *sel0* signals to the decoder ASIC. These PMOD pins from the FPGA board generate 0 and 3.3 V for low and high levels of the signal respectively. However, ASIC under test has been fabricated in 130-nm CMOS process with a supply voltage of 1.2 V and thereby, signal conditioning circuitry has been used for down converting these voltage levels to 0 and 1.2 V for low and high values, respectively.

To begin decoding, reset signal must be made low (using the switch) which initiates fetching of input LLRs from BRAMs at the FPGA clock rate (same rate at which the chip operates) and are fed to signal conditioning circuitry via PMOD connectors. Subsequently, the signal conditioned input-LLRs are fed to our

TABLE I
COMPARISON OF THE PROPOSED MAP DECODER WITH THE REPORTED WORKS

	[9]♣JSSC-05	[10]♣ASSCC-06	[11]✕TVLSI-11	[13]✕TCAS1-14	[12]♣TCAS1-12	[15]✕TCAS1-19	This work♣
Technology (nm)	180	130	130	130	180	40	130
Supply (V)	1.8	1.2	1.2	1.2	1.8	0.9	1.2
Area (mm ²)	8.7	1.96	1.28	2.12	3.0	2.8	1.28
Scaled Area (mm ²) [‡]	4.54♣	1.96◇	1.28◇	2.12◇	5.75♣	29.58‡	1.28◇
Max. Clock Freq. (MHz)	285	238	125	526	465	364	204
Total Power (mW)	330	528	97.83	NA	1860	240	92
Radix- <i>r</i>	<i>r</i> = 2	<i>r</i> = 16	<i>r</i> = 4	<i>r</i> = 2	<i>r</i> = 2	<i>r</i> = 2	<i>r</i> = 2/4/8
Throughput (Mbps)	27.6	952	500	526	465	364	201–605
Quantization (bits)	9	6	-NA-	7	10	10	8
η_α (bits/mm ²)	0.021 [†]	2.04	3.13	0.47	0.639 [†]	0.033 ^b	2.32
η_e (nJ/bit)	2.77 [§]	0.55	0.195	-NA-	0.927 [§]	12.38 ^ℒ	0.15

✕: Synthesized and Post-Layout Simulated Results; ♣: Measured Results from ASIC Chip.

‡: Scaled Area = (Area/*s*²) where *s* = Scaling Factor; ♣: *s* = (180/130); ◇: *s* = (130/130); ‡: *s* = (40/130).

η_α = Area Efficiency = Throughput / (Area × Max. Clock Frequency); †: Scaled Area Efficiency = $\eta_\alpha \times (180/130)^2$.

^b: Scaled Area Efficiency = $\eta_\alpha \times (40/130)^2$; η_e = Energy Efficiency = (Power / Throughput).

§: Scaled Energy Efficiency = $\eta_e \times (130/180)^2 \times (1.2/1.8)^2$; ℒ: Scaled Energy Efficiency = $\eta_e \times (130/40)^2 \times (1.2/0.9)^2$.

chip in every clock cycle for decoding, as shown in Fig. 13. Here, the decoder ASIC has been dc power supplied with 1.2 and 3.3 V for core and I/O pads respectively. Another power supply of 3.3 V from channel-3 is for the operational amplifiers used in the signal conditioning circuitry. Furthermore, the output pins of MAP decoder chip are connected to three channels of mixed signal oscilloscope (Agilent MSO-X 3054A) to view the final outcome, as shown in Fig. 13. The MAP-decoder ASIC has been operated in radix-8 mode and thereby, three bits are decoded in every clock cycle as shown in Fig. 14. Here, channel 1 of MSO displays the clock signal and channels 2–4 show the decoded bits. In addition, Monte Carlo testing is carried out at each SNR to obtain the BER from the decoder chip operating at radix-4 mode. These measured BER values are plotted against the simulated BER plots, as shown in Fig. 14. Here, the measured plot has a coding loss of 0.5 dB at a BER of 10^{-4} with respect to the simulated one.

D. Comparison

The measured results of our design have been compared with similar implementations from the literature, as listed in Table I. It shows the chip implementation results of Lee *et al.* [9] and Tang *et al.* [10]. Postlayout simulated results reported by Lin *et al.* [11] and Shrestha and Paily [13] are also included in Table I. Furthermore, it presents the comparison with recently reported work on low-routing complexity MAP decoder-architecture by Lin and Hsieh [15]. Our implementation is also compared with the measured results of MAP decoders with various states and their implementation

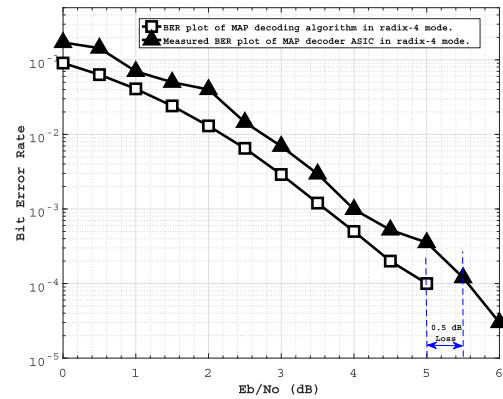


Fig. 14. Measured output waveform from the MAP-decoder ASIC and its BER performance comparison operation at radix-8 and radix-4 operating modes, respectively.

tradeoffs, reported by Studer *et al.* [12]. Table I shows that the proposed decoder is the only design that can operate in radix-2/4/8 modes consuming the least power and area among all.

It delivered higher throughput up to 605 Mb/s which is 39.8% higher than the currently reported work from [15]. For a fair comparison, we have computed two figure of merits: area efficiency (η_a) and energy efficiency (η_e), as shown in Table I. Here, η_e indicates the energy consumed by our chip to decode a single bit and η_a represents the number of hard decoded bits per mm^2 of the chip area. Table I shows that the proposed ASIC is most energy efficient design which delivers an efficiency of 0.15 nJ/bit. This value is 30% better than the best energy efficiency delivered by the reported work in [11]. However, our design delivers area efficiency of 2.32 bits/ mm^2 which is $1.35\times$ lower and $70\times$ higher than the implementations in [11] and [15], respectively, as shown in Table I.

V. CONCLUSION

This article demonstrated hardware implementation and testing of 2.35-mm² ASIC chip of MAP decoder with an energy efficiency of 0.15 nJ/bit. This decoder operates in multiple radix-2/4/8 modes and has achieved variable throughput between 201 and 605 Mb/s. In addition, we presented multi-radix MAP-decoding algorithm and its architecture design.

This new MAP decoder has special capability of operating with variable data-rates and it is essential in various wireless communication applications where energy efficiency is a significant parameter. It is to be noted that most of the wireless applications use hand-held devices and energy conservation is an essential factor to be optimized. Furthermore, MAP decoder is a primary kernel of widely adopted turbo and LDPC decoders. Hence, efficient design of such MAP kernel will enhance the overall design performances of these channel decoders. On the other side, MAP decoder can be used as an independent channel decoder where near Shannon-limit performance is not a necessary criterion.

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